Dense Phantom Boundary And Action Obstruction

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Summary

Pass 84 studies the dense quotient

$$\mathbb{R} \to \Sigma \to \epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}, \qquad \Sigma = (\mathbb{R} \times \widehat{\mathbb{Z}})/\mathbb{Z}.$$

The topological quotient ϵ is indiscrete, so it has no nontrivial continuous maps to Hausdorff targets. The finite-prime phantom survives only as a solid derived boundary.

Main Claims

- **Indiscrete quotient.** Since $\mathbb{Z}$ is dense in $\widehat{\mathbb{Z}}$, every nonempty open subset of $\widehat{\mathbb{Z}}$ saturates to all of $\widehat{\mathbb{Z}}$ under addition by $\mathbb{Z}$. Hence $\widehat{\mathbb{Z}}/\mathbb{Z}$ is indiscrete and its Hausdorff reflection is 0.
- **No continuous translation action.** Any continuous homomorphism from ϵ to the compact Hausdorff solenoid Σ is zero. Therefore the Borel unipotent $U = \epsilon$ does not act by nontrivial continuous translations on Σ.
- **Derived Weyl replacement.** In the solid derived category,

  $$D\epsilon \simeq \mathbb{Q}[-1], \qquad \mathrm{Ext}^1_{\mathrm{Solid}}(\epsilon, \mathbb{Q}) \simeq \mathbb{Q}.$$

  The missing Weyl flip $\epsilon \to \mathbb{Q}$ is replaced by the degree-1 finite-adele extension

  $$0 \to \mathbb{Q} \to \mathbb{A}_f \to \epsilon \to 0.$$

Machine Verification

`code/scripts/check-pass84.py` produced `artifacts/reports/pass84-dense-phantom-boundary-action-check.json` with overall PASS. It checks finite saturated-open shadows, finite continuous maps from an indiscrete quotient to discrete targets, character descent, and the absence of finite degree-0 Weyl shadows into $\mathbb{Q}$.

Next Use

Pass 85 should build an explicit two-term complex model comparing $[\mathbb{Z} \to \widehat{\mathbb{Z}}]$, $[\mathbb{R} \to \Sigma]$, and $[\mathbb{Q} \to \mathbb{A}_f]$, then identify which quasi-isomorphisms preserve the Borel shear class.